Machine Learning for Tabular Data

Fundamentals of Artificial Intelligence

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Supervised Learning in Mathematical Terms

map input to output: $y = f(\boldsymbol{x})$

random variables Y and $\boldsymbol{X} = (X_1, X_2, \cdots, X_p)$ ← usually high-dimensional

training (e.g., curve fitting / parameter estimation):

fit data set of $(y_i, \boldsymbol{x}_i)$ pairs → minimize deviations between y and $f(\boldsymbol{x})$

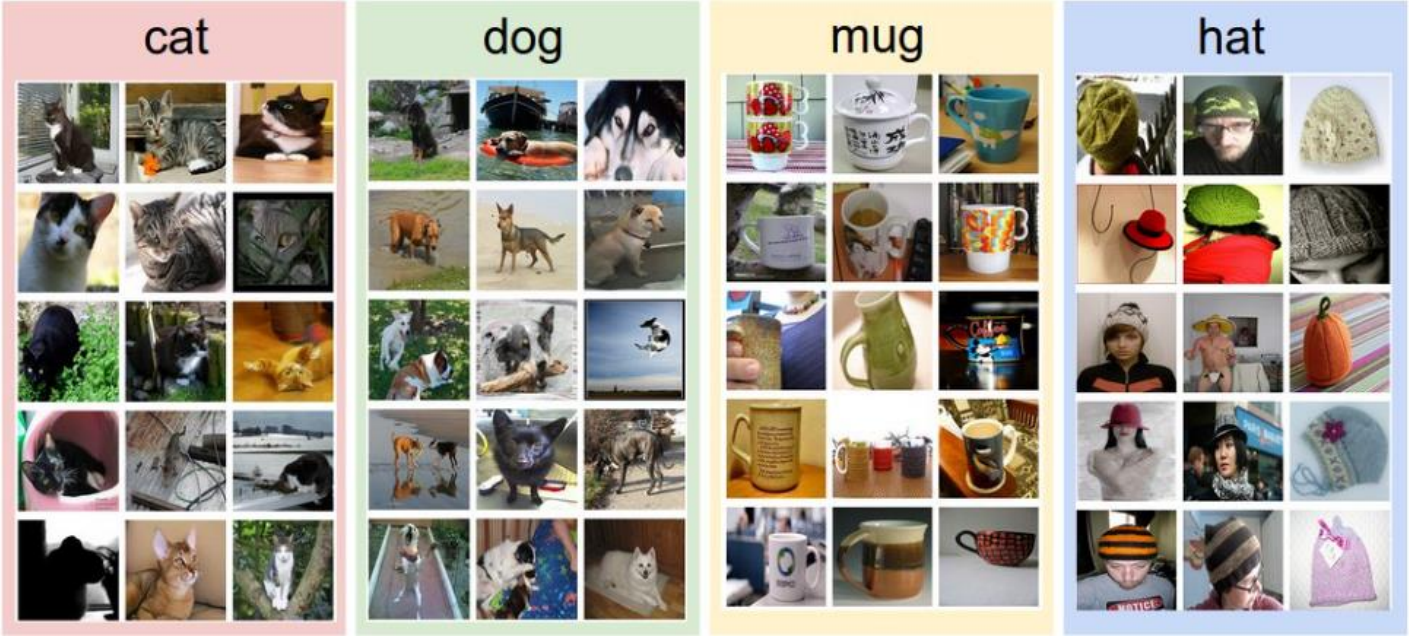
discriminative models: predict conditional density function $p(y|\boldsymbol{x})$

as opposed to generative models: predict $p(y, \boldsymbol{x})$ (or just $p(\boldsymbol{x})$) → $\boldsymbol{x}$ not given, more difficult

Example: Image Classification

training data set:

labels:



test data with learned classifier:

$f($ $) =$ "Cat"

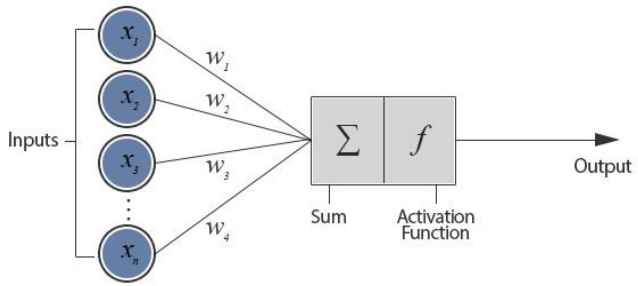
$f($ $) =$ "Dog"

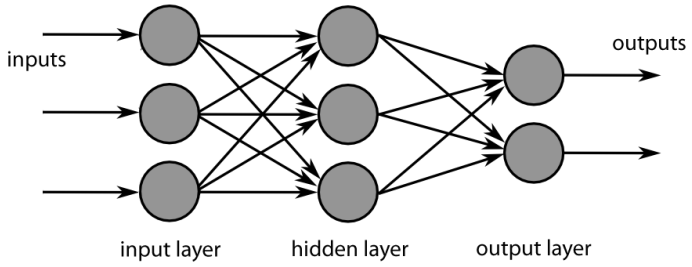
Algorithmic Families of Supervised Learning

parametric models

linear models

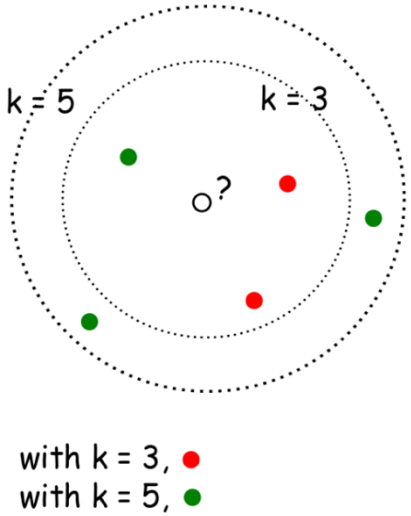
neural networks





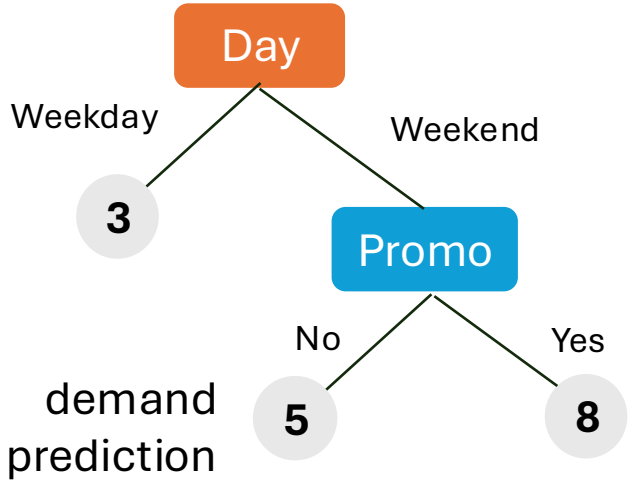
non-parametric models: similarity-based learning (analogies)

nearest-neighbor-style/local methods



kernel machines (e.g., support-vector machines)

decision trees: rule learning



often used in ensemble methods

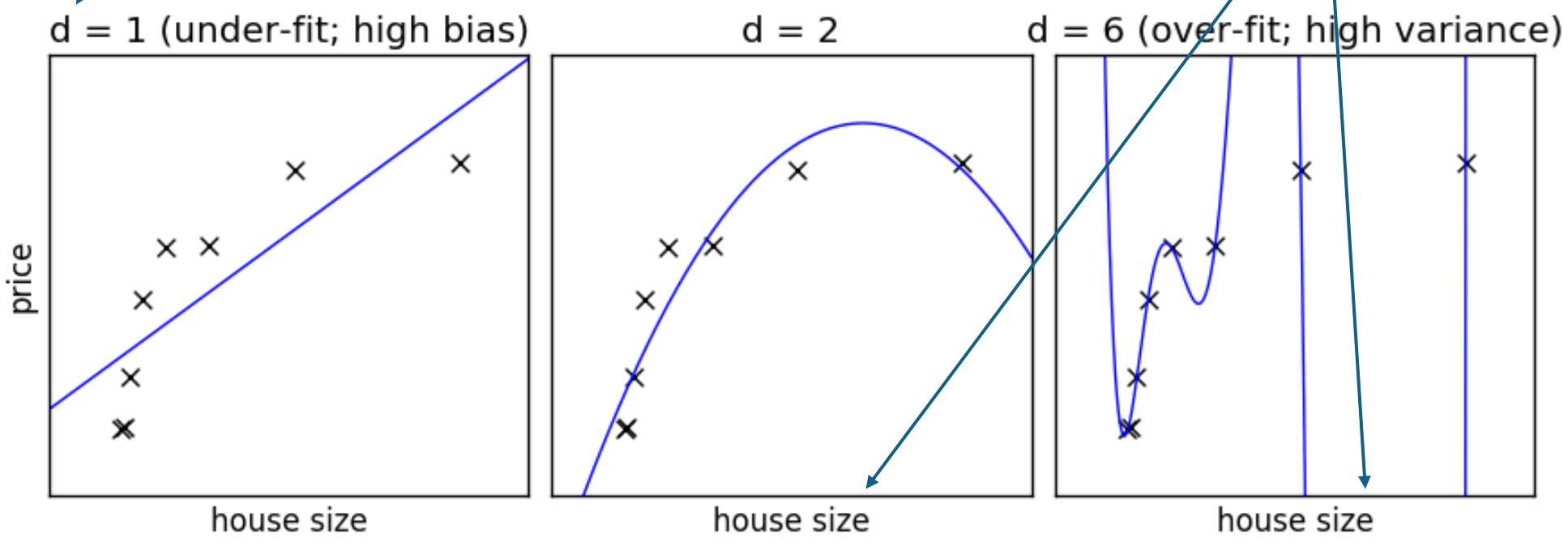
- bagging: random forests
- boosting: gradient boosting

Under- and Over-Fitting

example: polynomial fit

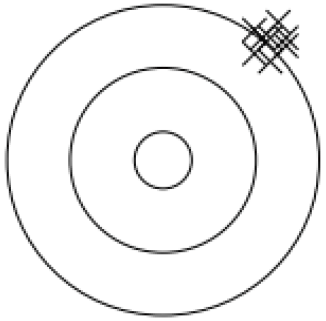
need to predict new samples (not in training data set): interpolation

degree of fitted polynomial

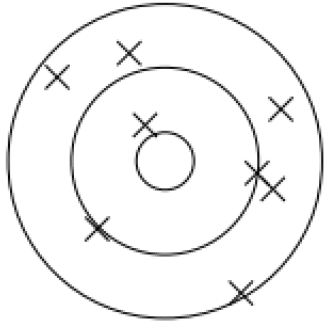


Bias, Variance, Irreducible Error

think of fitting ML algorithms as repeatable processes with different data sets



bias
due to too simplistic model
(same for all training data sets)
"underfitting"



variance
due to sensitivity to specifics (noise)
of different training data sets
"overfitting"

irreducible error (aka Bayes error):

inherent randomness (target generated from random variable)

→limiting accuracy of ideal model

Bias-Variance Tradeoff

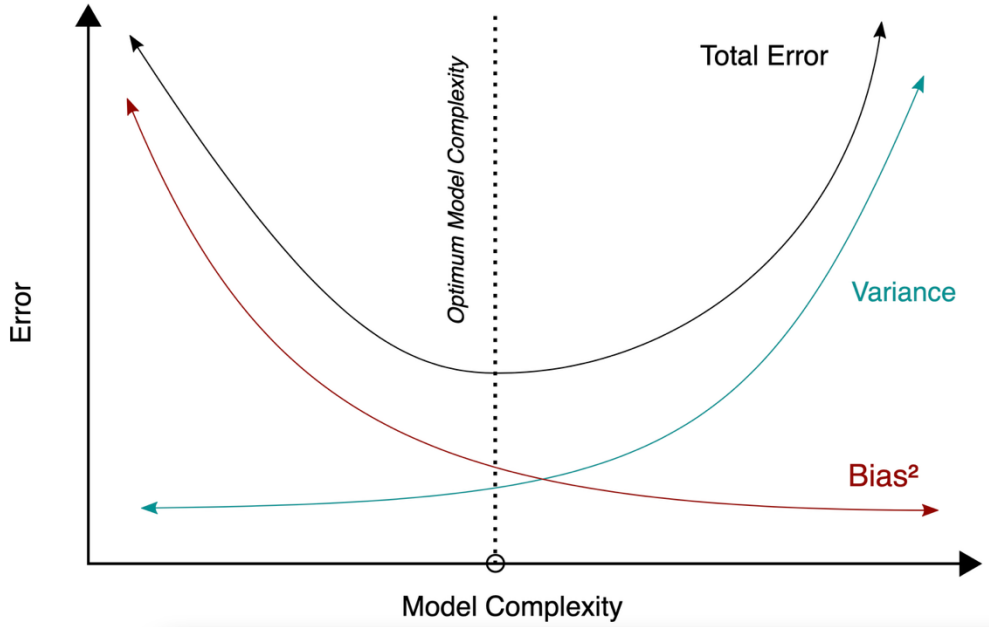
models of higher complexity have lower bias but higher variance

(given the same number of training examples)

generalization error follows U-shaped curve: overfitting once model complexity (number of parameters) passes certain threshold

overfitting: variance term dominating test error

→ increasing model complexity increases test error



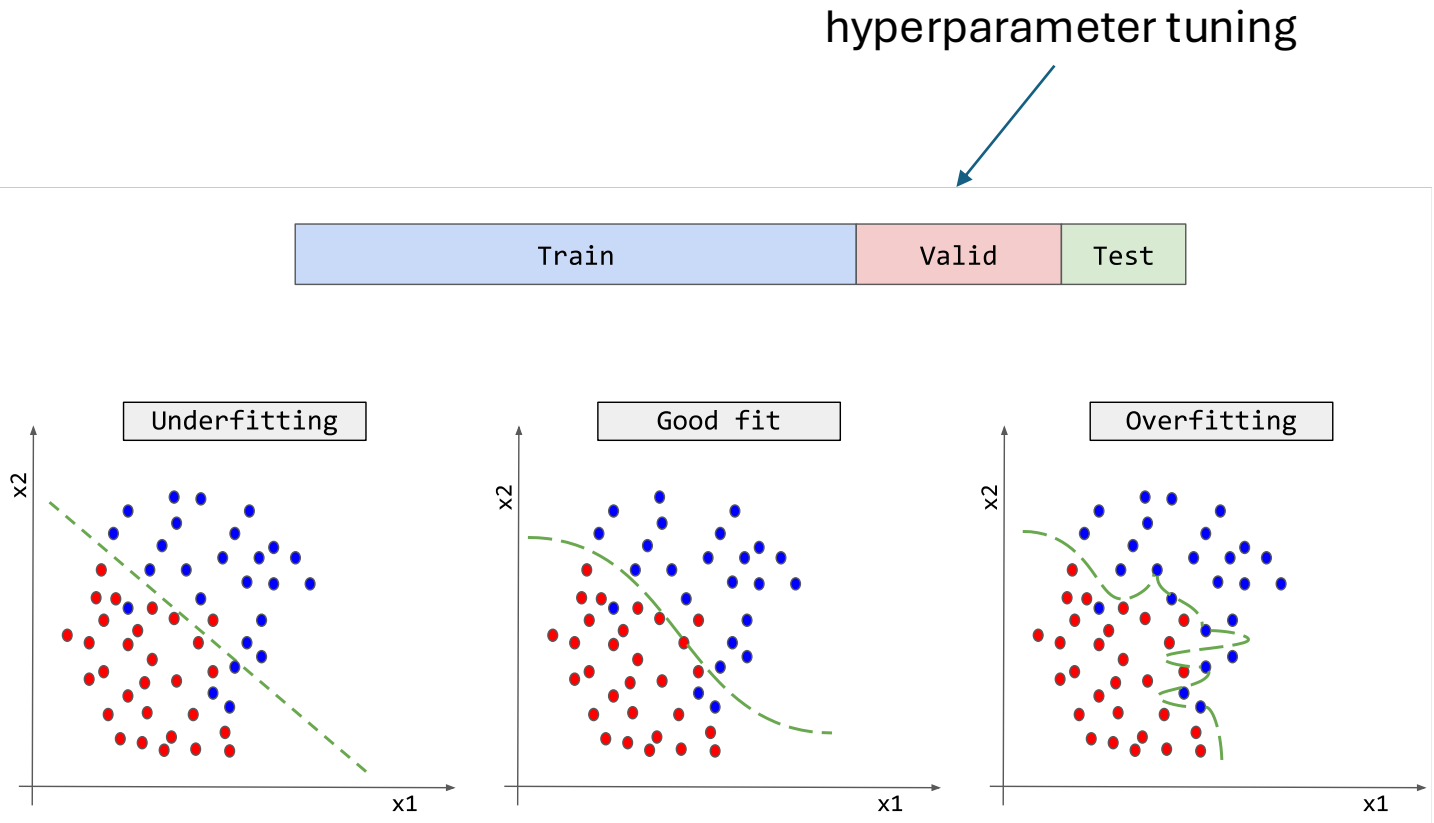
but beware: training error keeps decreasing with more complex model ...

Hyperparameters

model complexity often controlled via hyperparameters (parameters not fitted but set in advance)

examples:

- degree of fitted polynomial d
- number of considered nearest neighbors k
- maximum depth of decision tree
- number of hidden nodes in neural network layer



Train-Dev-Test vs. Model fitting

Generalization

core of ML: **empirical risk minimization** (training error) as proxy for minimizing unknown population risk (test error)

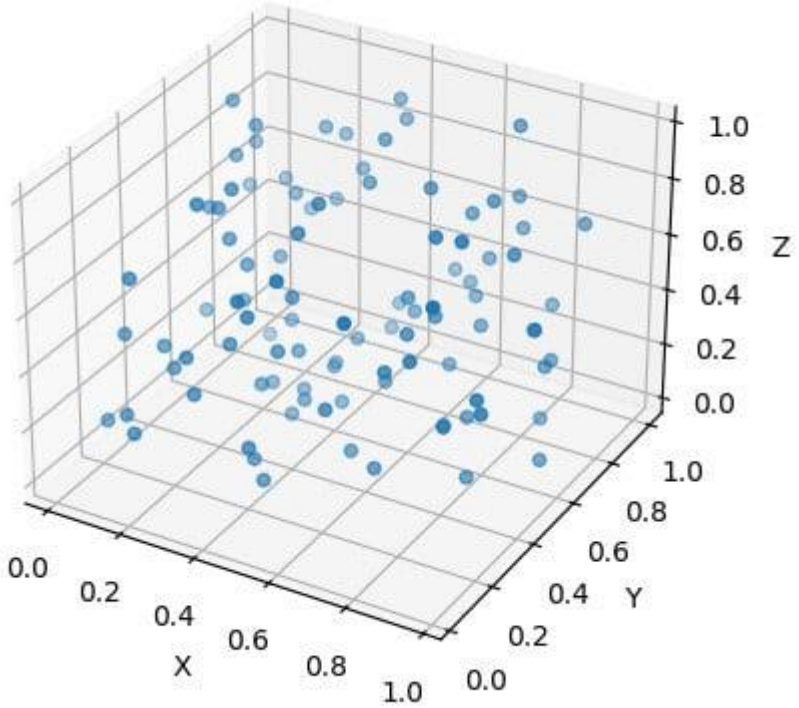
generalization gap: difference between test and training error

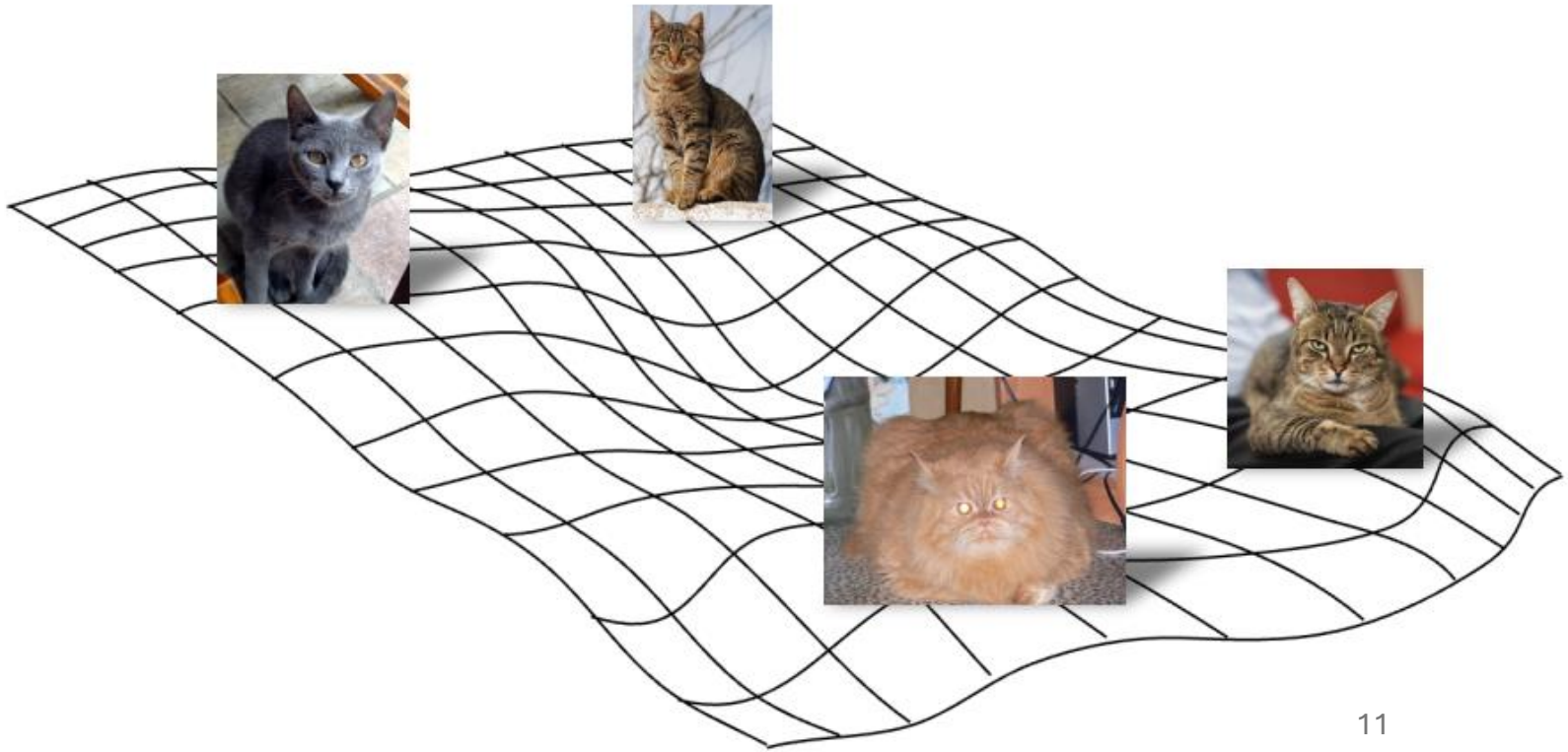
curse of dimensionality: typically, many features (dimensions)

→ lots of data needed to densely sample volume

Manifold Hypothesis

luckily, reality is friendly: most high-dimensional data sets reside on lower-dimensional manifolds → enabling effectiveness of ML





Linear Regression

fit:

$$y_i = \overbrace{b + \sum_{j=1}^{p} w_j \, x_{ij}}^{f(\boldsymbol{x}_i)} + \varepsilon_i$$

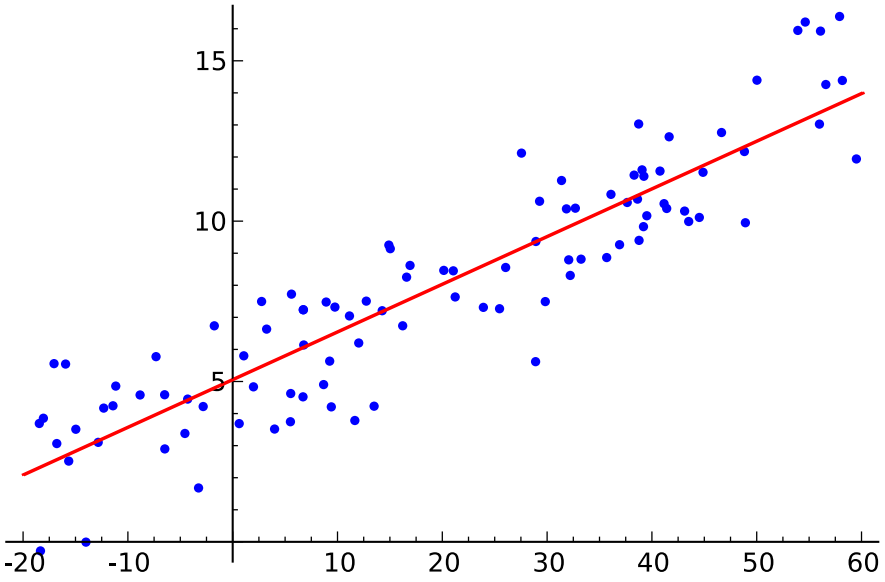
Gaussian

predict:

$$\hat{y}_i = E[Y|\boldsymbol{X} = \boldsymbol{x}_i] = f(\boldsymbol{x}_i) = b + \sum_{j=1}^{p} w_j \, x_{ij}$$

$$p(y|\boldsymbol{x}_i) = \mathcal{N}(y; \hat{y}_i, \sigma^2)$$

Gaussian mean variance

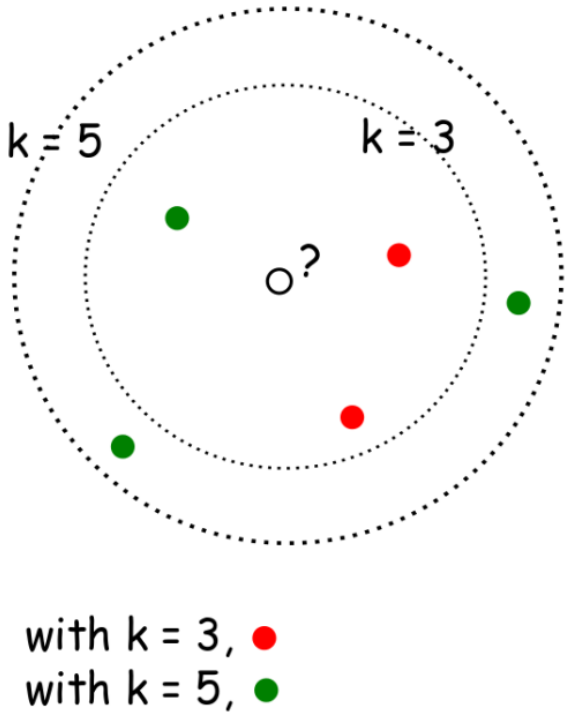


parameter estimation:
e.g., least squares

parameters to be estimated: $b, \boldsymbol{w}$
$\rightarrow \sigma^2 = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - f(\boldsymbol{x}_i)\right)^2$

k-Nearest Neighbors (kNN)

- local method, instance-based learning
- non-parametric
- distance defined by metric on $\boldsymbol{x}$ (e.g., Euclidean)



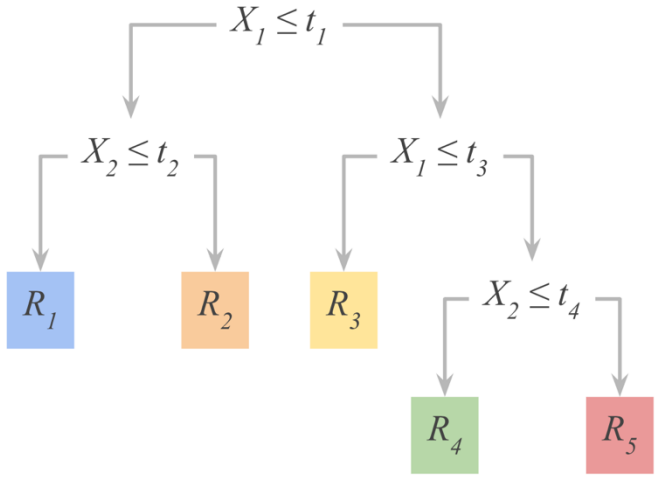
regression:

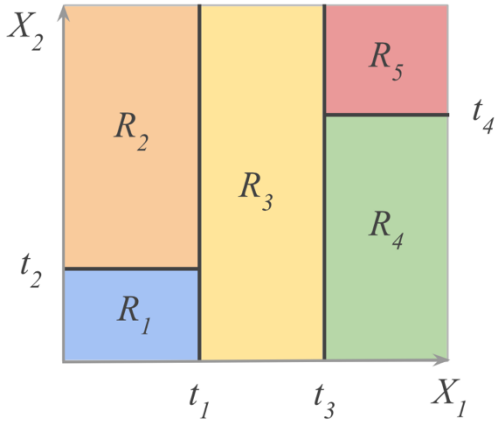
$$f(\boldsymbol{x}_0) = \frac{1}{k}\sum_{j=1}^{k} y_j$$

with j running over k nearest neighbors of $\boldsymbol{x}_0$

low k: low bias but high variance
high k: low variance but high bias

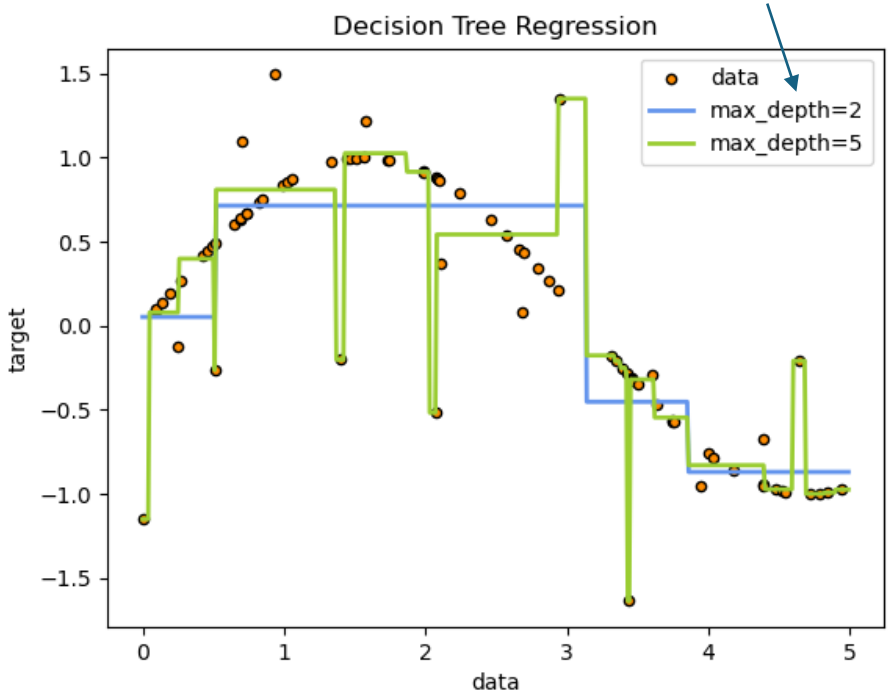
Decision Trees





- non-parametric learning of simple decision rules
- usually, binary trees
- axis-parallel decision boundaries (box-shaped regions in feature space)
- fit constant in each box
  - classification: majority class of target
  - regression: average of target
- fully explainable models

hyperparameter



scikit-learn

Ensemble Learning

idea: combine several individual models (often of the same type) to form an ensemble model with better predictive performance than each of its constituents

→ learning in several different ways from the training data

- the trainings of the individual constituent models are (kind of) independent
- outputs of individual models are combined (e.g., averaged)

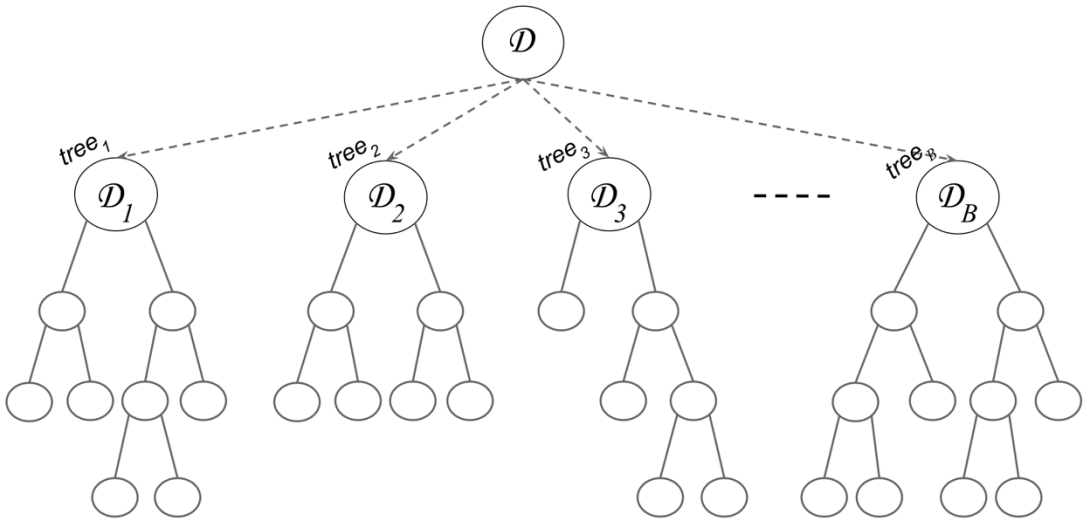
Bagging (Bootstrap Aggregating)

idea: train individual models of ensemble on random subsets of the training data (random sample with replacement)

- reduces variance of ensemble model compared to individual models
- but not bias → use low-bias base models

individual models combined by averaging (regression) or majority voting (classification) → committee of models

example with decision trees:



source

Random Subspace Method

aka feature bagging

idea: train different members (models) of an ensemble method on random subsets of all features (random sample with replacement)

→ reduce correlation between ensemble members

individual models combined by averaging (regression) or majority voting (classification) → committee of models

Random Forests

ensemble method using

- decision trees as individual models (usually, rather large, low-bias decision trees)
- combination of bagging and random subspace method (features sampled at each node in the trees)

compared to individual decision trees, random forests

+ reduce variance → improve accuracy

- lose explainability

one of the most popular off-the-shelf ML algorithms (often, good performance with little hyperparameter tuning and data preparation)

Boosting

idea: sequentially learn and combine several “weak” learners (such as small decision trees, but in principle any ML algorithm) to construct a “strong” one

→ gradually (in a greedy fashion) reducing bias of ensemble model

in simple terms:

building a model from the training data, then creating a second model that attempts to correct the errors from the first model, ...

→ each subsequent weak learner is forced to concentrate on the examples that are missed by the previous ones in the sequence

not a committee of models:

typically, use simple, high-bias methods as individual models (e.g., small decision trees)

most prominent: AdaBoost, Gradient Boosting

Forward Stagewise Additive Modeling

boosting can be understood as fitting an additive expansion in a set of basis functions (e.g., decision trees): $f(\boldsymbol{x}) = \sum_{m=1}^{M} \gamma_m h_m(\boldsymbol{x})$

loss minimization of $f(\boldsymbol{x})$ computationally expensive → sequentially add and fit individual basis functions without changing already fitted ones

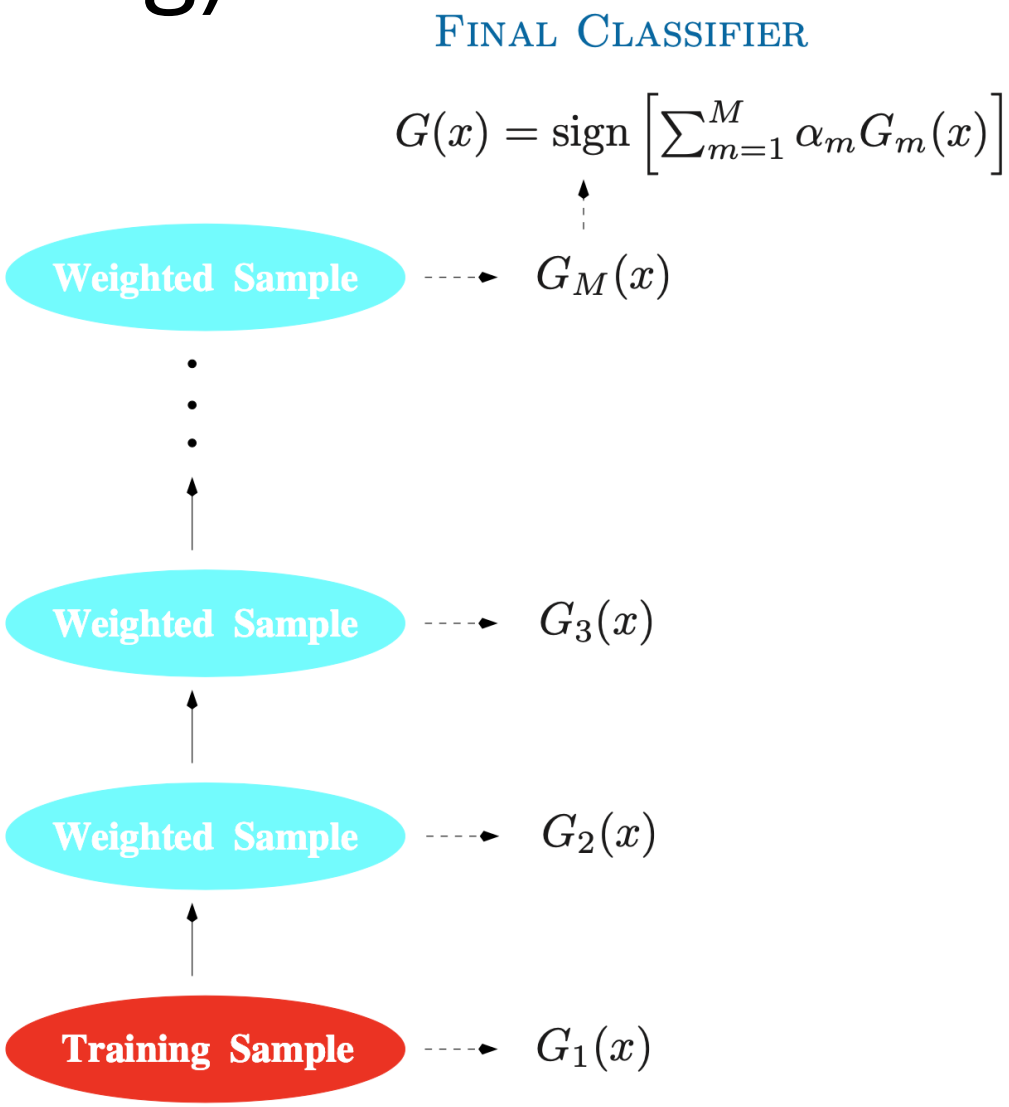
boosting algorithm:

for $m = 1$ to M:

1. compute $\underset{h_m, \gamma_m}{\operatorname{argmin}} \sum_{i=1}^{n} L\big(y_i, f_{m-1}(x_i) + \gamma_m h_m(x_i)\big)$ (initialized with $f_0(\boldsymbol{x}) = 0$)
2. set $f_m(\boldsymbol{x}) = f_{m-1}(\boldsymbol{x}) + \gamma_m h_m(\boldsymbol{x})$

AdaBoost (Adaptive Boosting)

- iteratively train weak classifiers on weighted data
- emphasizing misclassified points (weights increased for incorrect predictions, decreased for correct ones)
- combine them into weighted vote (forward stagewise additive modeling)



Gradient Boosting

cost function

loss function

define $\mathcal{L}(f_m) = \sum_{i=1}^{n} L\big(y_i, f_{m-1}(x_i) + \gamma_m h_m(x_i)\big)$

in forward stagewise additive modeling:

$$f_m(\boldsymbol{x}) = \sum_{m=1}^{M} \gamma_m h_m(\boldsymbol{x})$$

$$\underset{h_m, \gamma_m}{\operatorname{argmin}} \sum_{i=1}^{n} L\big(y_i, f_{m-1}(x_i) + \gamma_m h_m(x_i)\big) \quad \text{for each stage}$$

→ can be hard to solve for general loss function L

idea: approximation of $\mathcal{L}$ by its Taylor expansion

gradient g at x_i:
constant for this iteration

$$\mathcal{L}(f_m) \approx \mathcal{L}(f_{m-1}) + \sum_{i=1}^{n} h_m(x_i) \cdot \left[\frac{\partial L(y_i, f(x_i))}{\partial f(x_i)}\right]_{f=f_{m-1}}$$

→ decreased by going a step of size γ_m (potentially optimized via line search) in the direction $h_m(\boldsymbol{x}) \approx -g$ (functional gradient descent)

→ find h_m by fitting a regression model with targets $-g$ (typically decision tree regressors of fixed size: gradient-boosted decision trees)

works like this for both regression and classification tasks (just different loss functions)

Decision Tree Sizes for Boosting

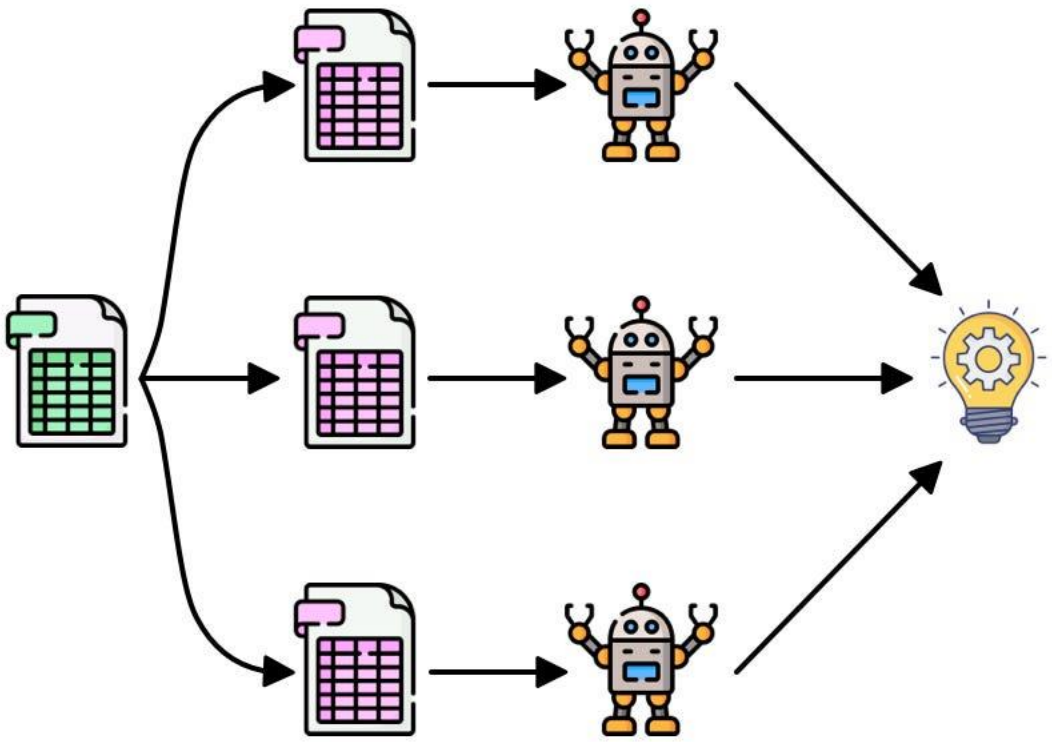
consider degree to which features interact with each other in given data set

- decision trees with single split (aka decision stumps): covering no interaction effects, just main effects of individual features
- decision trees with two splits: covering second-order interactions
- decision trees with three splits: covering third-order interactions

(usually, interaction order not much higher than ~5 in real-world data sets)

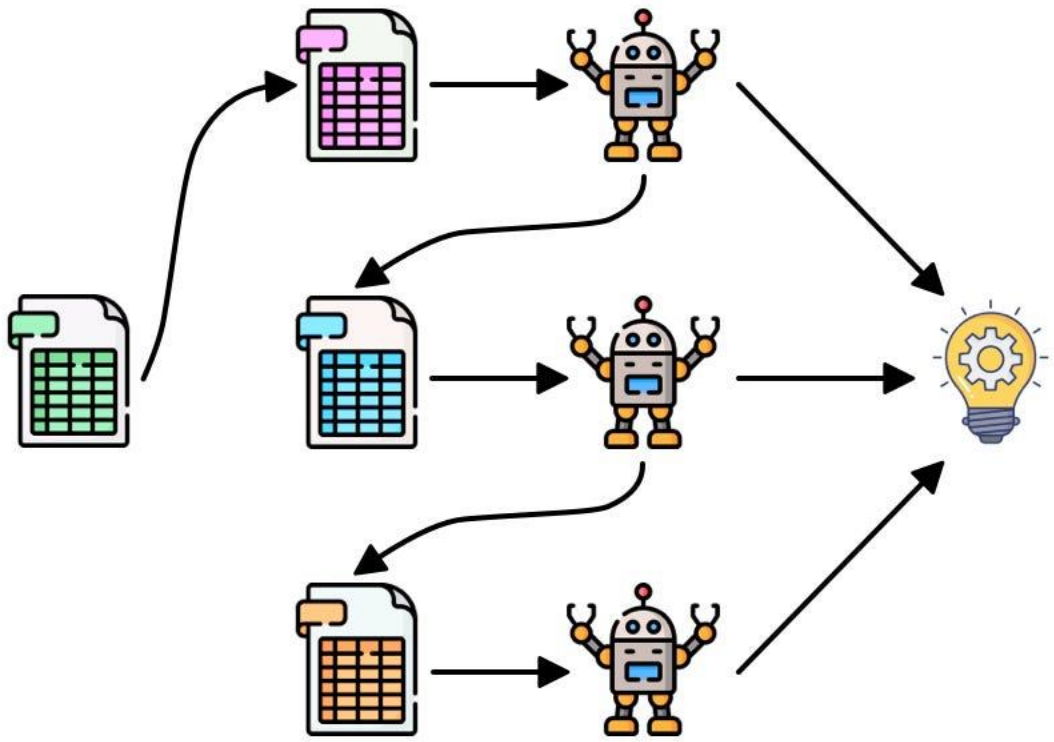
why boosting often works better than single large, low-bias model:
uncorrelated learners, each focusing on a specific aspect of the data

Bagging



Parallel

Boosting



Sequential